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Pioneering Digitalization in Power Sector

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AMI Smart Meter Rollout to Smart Data Operations

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INDIA'S SMART METER JOURNEY

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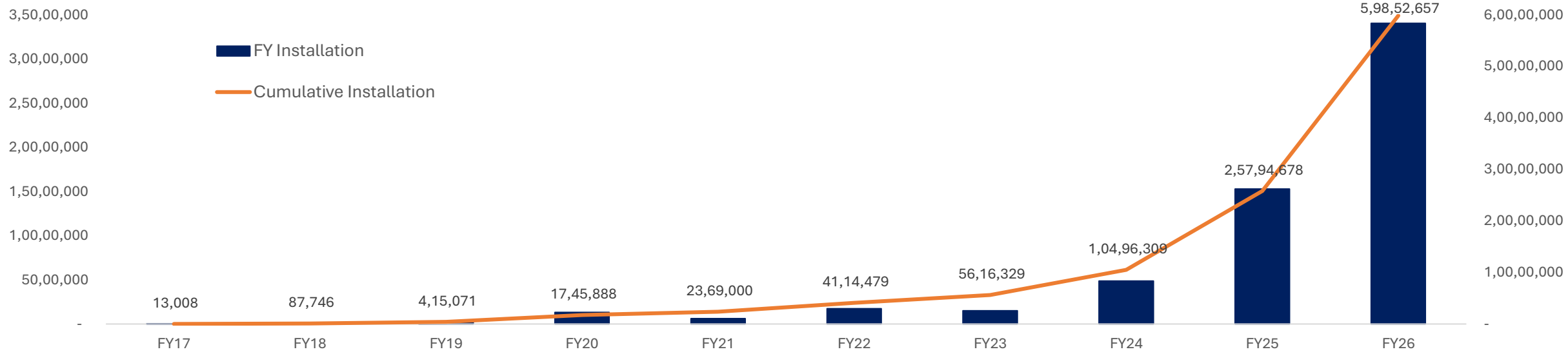
Smart Meters Sanctioned*	22.98 Cr	Tenders Awarded*	15.58 Cr	Total Smart Meters Installed*	5.98 Cr



- India today is **among the fastest smart meter deployment** markets globally, both in scale and speed even at the current installation rate, already it is the fastest ever rollout achieved around the globe.
- This is one of the largest grass root level digitalization programme in the country and Honorable Prime Minister has appreciate the Smart Metering Programme in Assam.
- RDSS has strongly reinforced the **Make in India** agenda, Indigenous meter manufacturing, Domestic communication modules, Indian HES, MDMS, and analytics platforms
- Programme is slowly gaining trust of the Consumers, all progressive state governments are now boldly promoting Smart Metering

India has solved the scale problem. The next challenge is extracting value from data.

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THE INFLECTION POINT

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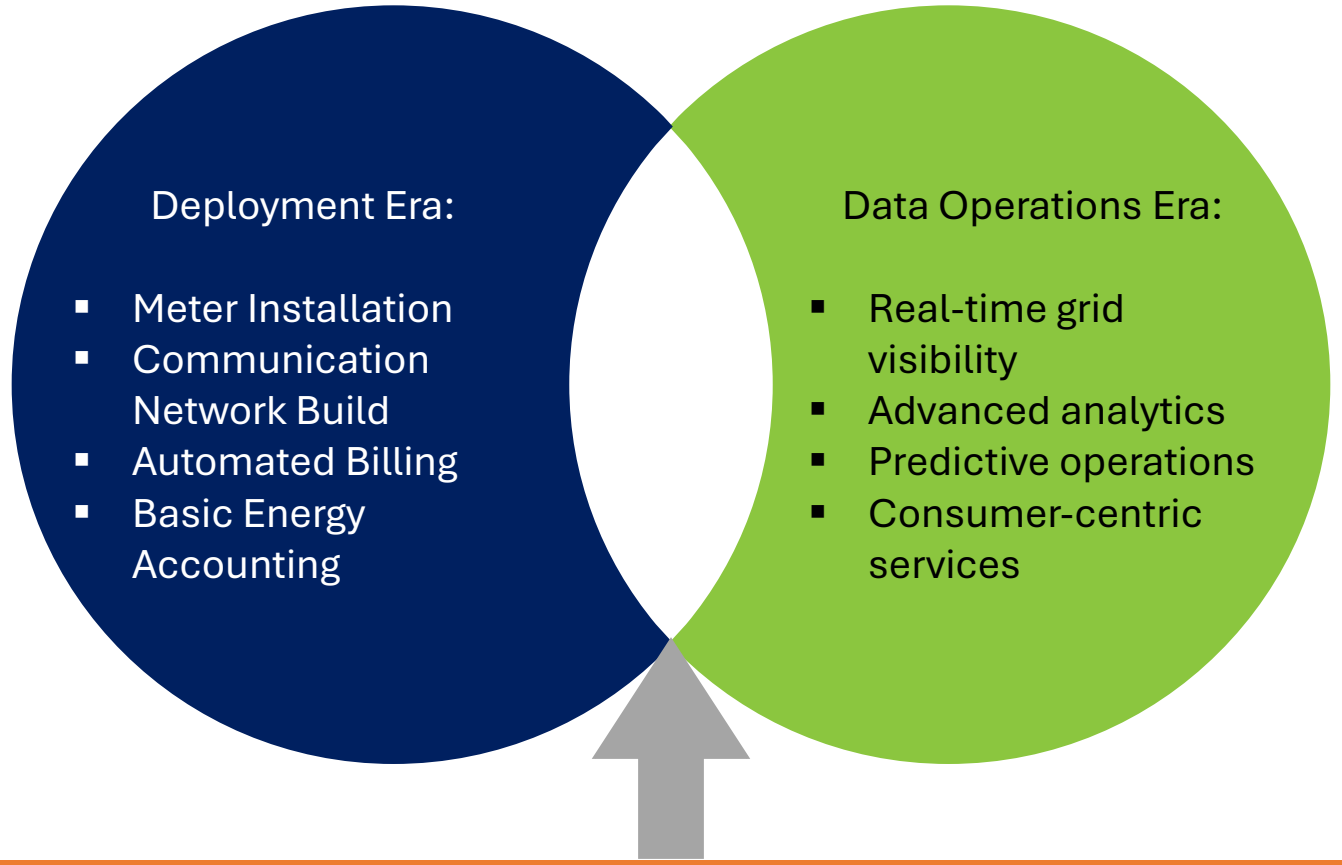
The sector is moving from meter deployment to digital utility operations



We are at this inflection point of Smart metering journey in the country

Smart meters are not endpoints, they are data infrastructure for digital DISCOMs

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GRID INTELLIGENCE: THE DATA OPPORTUNITY

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Smart meters create one of the richest datasets in the power sector

Millions of meters, billions of data points daily

Smart meter generates:

- 30-minute interval consumption data
- Event logs
- Voltage & power quality parameters
- Load curves
- Tamper alerts

When integrated with:

- SCADA
- GIS
- Billing systems
- Outage management

Grid Intelligence

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Key Focus Area

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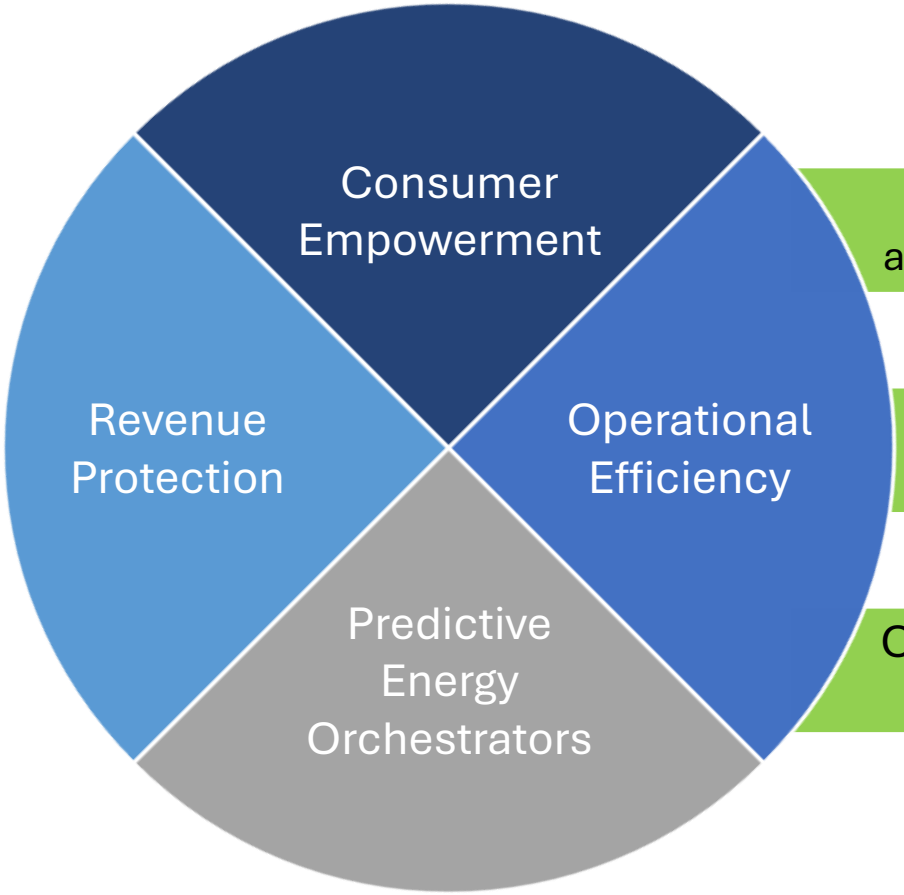

Advanced Distribution Management Systems (ADMS) with AI analytics. Provides grid stability analysis and outage prediction

Monitor transformers, feeders and substations using IoT sensors and prevent equipment failure.

Computer vision models analyze drone images of power infrastructure to detect damages.

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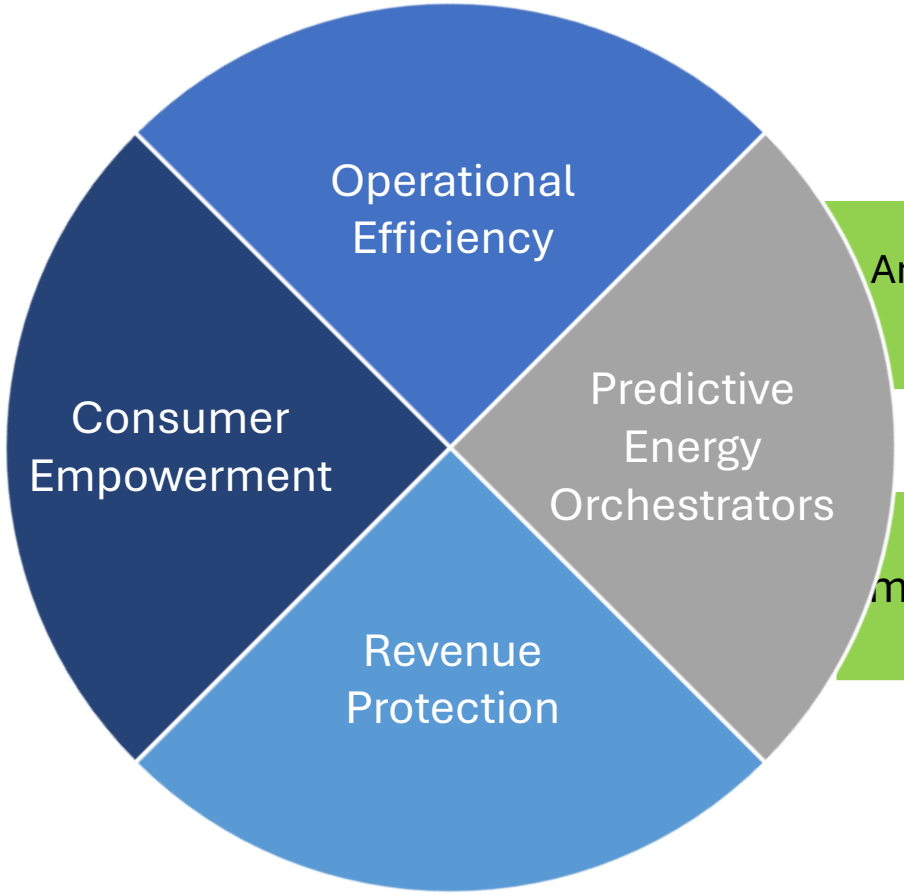

Analyze transformer load, temperature and historical fault data and predict potential transformer failures before breakdown.

AI algorithms optimize demand response and energy management for utilities and large consumers for peak load reduction.

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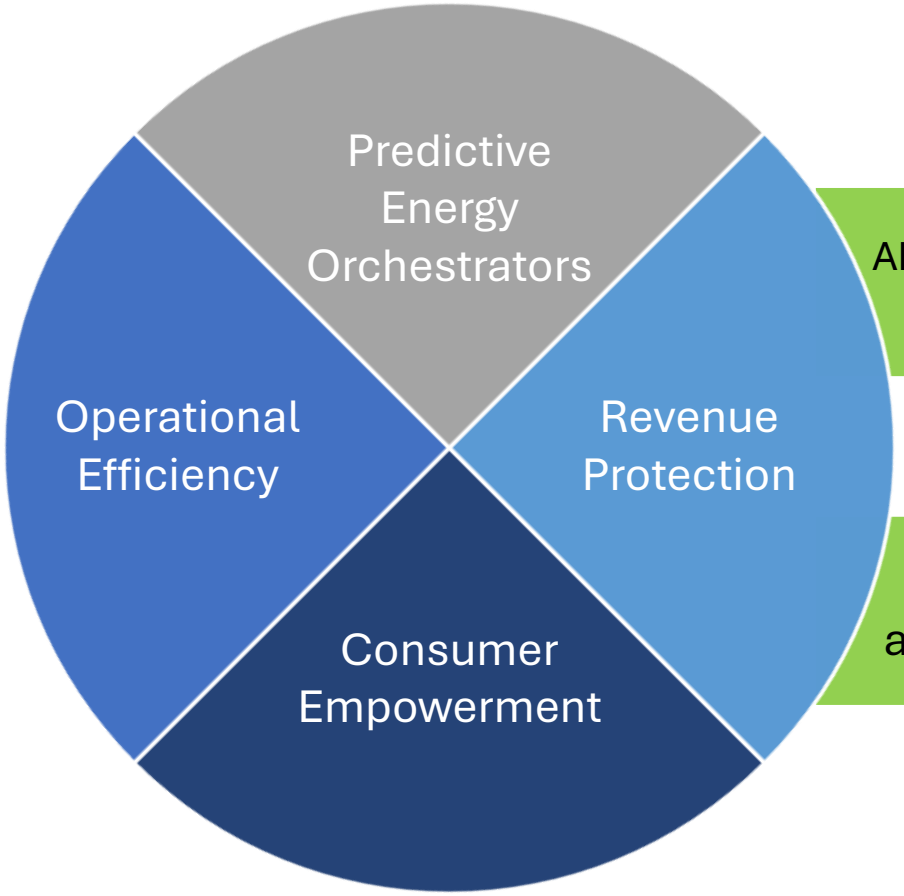

AI Energy Theft Detection: AI models analyze smart meter interval data to detect theft patterns

Smart meter data analytics for consumption pattern analysis, anomaly detection, feeder-level energy auditing

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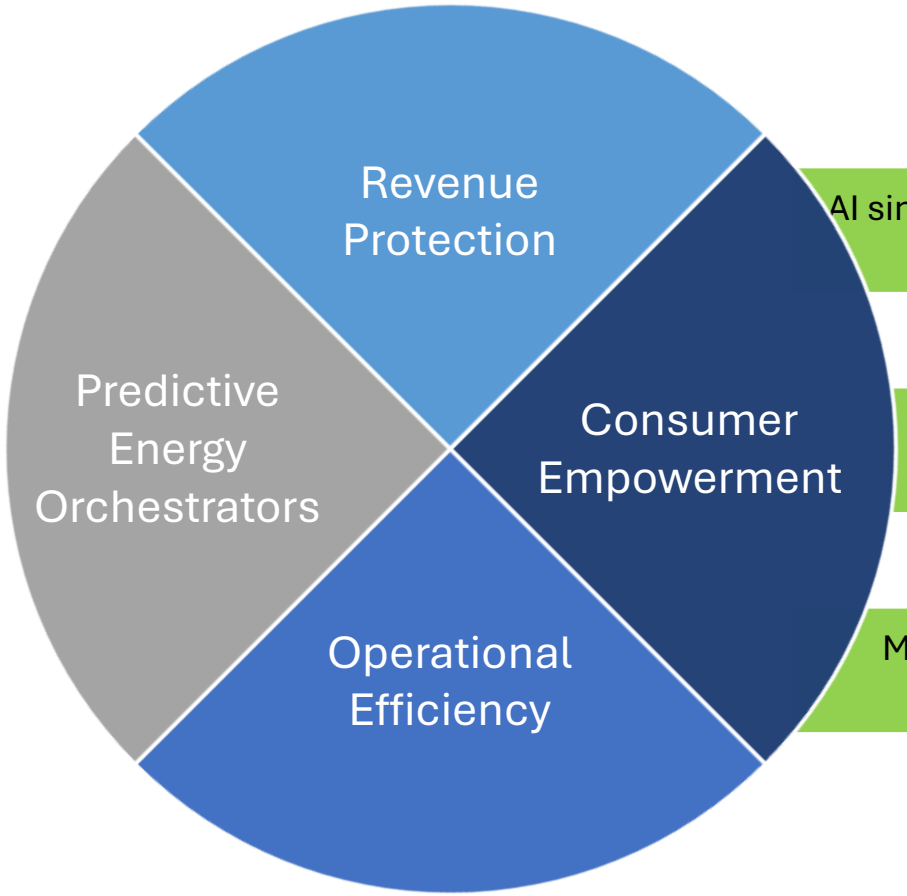

AI simulations optimize distribution network expansion planning. Enabling P2P trading, Integration of EV load growth and rooftop solar penetration.

Appliance level Energy Disaggregation: Targeted demand-side management

ML models segment consumers based on consumption behavior. Enables targeted tariffs & demand-side programs.

India's distribution sector has evolved from basic electrification to digital metering to data-driven utilities to AI-enabled grid intelligence, with smart meters and analytics forming the backbone of the future intelligent grid

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WAY FORWARD ON THE DIGITAL JOURNEY

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- Strengthen Smart Meter Operations Centers and data quality
- Integrate AMI with SCADA, GIS and outage systems
- Move from installation KPIs to outcome KPIs
- Build data analytics capability
- Use smart meter data for Demand Side Management and consumer engagement

Deployment is the **first step**
Data utilization will determine **sector transformation**

The next phase of India's power sector reform will be defined not by how many smart meters we install — but by how intelligently we use the data they generate

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Thank You

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